Bayesian and sampling-based cortical computation

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This guide connects the neural-Bayesian literature to our PING project. The central question is not merely whether cortical behaviour can look Bayesian. It is whether recurrent excitatory/inhibitory (E/I) dynamics can represent uncertainty by sampling, and whether the oscillatory and transient regimes we measure are computationally useful rather than incidental.

1. Papers and their relation to the project

1. **Pouget, Dayan & Zemel (2003), population-code foundations.** A broad account of inference with population activity [**1**]. It supplies the coding language needed to distinguish a distribution represented by a population from a single decoded estimate. For this project, it is the conceptual baseline: before asking whether PING dynamics sample a posterior, we need to state what information the population activity carries and what a downstream readout could recover.
2. **Ma, Beck, Latham & Pouget (2006), probabilistic population codes.** This is the canonical parametric alternative to neural sampling [**2**]. Instantaneous population rates encode the parameters of an exponential-family distribution, and cue combination becomes addition of population activity. It gives us a competing hypothesis: uncertainty may be encoded in population gain rather than in temporal variability. Any sampling claim from our network should make predictions that separate these accounts.
3. **Fiser, Berkes, Orbán & Lengyel (2010), the sampling hypothesis.** This review develops the proposal that spontaneous and evoked cortical activity are samples from an internal generative model [**3**]. It connects trial-to-trial variability to probabilistic representation. For our project, it motivates treating fluctuations as signal-bearing dynamics and asking whether the stationary activity distribution, rather than only the mean firing rate or oscillation frequency, matches a target distribution.
4. **Aitchison & Lengyel (2016), E/I circuits as Hamiltonian samplers.** This paper is the closest theoretical bridge to our model [**4**]. It assigns different computational roles to excitatory and inhibitory populations and argues that their coupled dynamics can accelerate sampling. It should inform which E/I variables we analyse, how phase-lagged activity might act like position and momentum, and why balanced oscillatory dynamics could improve exploration rather than simply stabilize the circuit.
5. **Orbán, Berkes, Fiser & Lengyel (2016), neural variability as sampling evidence.** This paper tests sampling-based predictions in visual cortex [**5**]. It is useful for translating the theory into empirical diagnostics, especially the relation between stimulus-

dependent uncertainty, response variability, and the distribution of population activity. Our analogue is to test whether trained PING networks change their variability in the way the represented posterior requires.

6. **Echeveste, Aitchison, Hennequin & Lengyel (2020), optimized recurrent samplers.** Recurrent networks optimized for sampling develop cortical-like transients and variability [**6**]. This is the most direct precedent for training a recurrent circuit toward a sampling objective. It motivates comparing task performance with mixing, autocorrelation, transient amplification, and stability, including our Δt-stability diagnostics. A network that fits the task but mixes poorly is not yet a convincing sampler.
7. **Padamsey, Katsanevaki, Dupuy & Rochefort (2022), precision under metabolic constraint.** Food scarcity reduces cortical coding precision and energy use [**7**]. This adds a cost axis to the project. If spike rate, inhibitory recruitment, or oscillatory precision has a metabolic price, the best circuit may trade posterior precision against energy rather than maximize accuracy without constraint.

2. Reading order and guide

Read the papers in four passes. The order below moves from representation, through the sampling hypothesis, to circuit implementation and finally energetic constraint.

1. **Establish the representational alternatives: Pouget et al. (2003), then Ma et al. (2006).** Ask what object neural activity represents, how uncertainty is encoded, and what operations a downstream circuit can perform. Keep the distinction between *encoding a distribution's parameters* and *producing samples from a distribution* explicit.
2. **Learn the sampling claim: Fiser et al. (2010), then Orbán et al. (2016).** Ask what counts as a neural sample, over what time or trial ensemble the distribution is defined, and which observations distinguish meaningful sampling variability from ordinary noise.
3. **Study the E/I mechanism: Aitchison & Lengyel (2016), then Echeveste et al. (2020).** Track the mapping from mathematical sampler variables to excitatory and inhibitory activity. Then look for measurable consequences in our networks: phase relations, autocorrelation time, mixing, transient amplification, stationary distributions, and sensitivity to the integration step Δt.
4. **Add the resource constraint: Padamsey et al. (2022).** Ask whether coding precision, firing cost, and inhibitory stabilization can be treated as a joint objective. This paper is best read last because it changes the optimization question from "what is the most accurate code?" to "what precision is worth its metabolic cost?"

On a first pass, read abstracts, figures, and discussion. On a second pass, reconstruct the representation and objective in equations. On a third pass, extract one table with four columns for each paper: represented quantity, circuit mechanism, observable prediction, and corresponding diagnostic in our PING model.

3. Terms and background to know

Bayesian fundamentals

A **generative model** is a forward account of how an unobserved cause θ produces an observation x. **Recognition** or **inference** reverses that account and estimates the cause from the observation. Bayes' rule is

$$p(\theta \mid x) = \frac{p(x \mid \theta)p(\theta)}{p(x)} \propto p(x \mid \theta)p(\theta).$$

- θ is the latent or unobserved cause.
- x is the observation.
- $p(\theta)$ is the **prior**, the distribution before observing x.
- $p(x \mid \theta)$ is the **likelihood**, the probability of x under a proposed cause.
- $p(\theta \mid x)$ is the **posterior**, the updated distribution over causes.
- $p(x) = \int p(x \mid \theta)p(\theta)\,\mathrm{d}\theta$ is the **evidence** or marginal likelihood, which normalizes the posterior.

Marginalisation integrates out variables that are not of interest. For a target θ_1 and nuisance variable θ_2,

$$p(\theta_1 \mid x) = \int p(\theta_1, \theta_2 \mid x)\,\mathrm{d}\theta_2.$$

- θ_1 is the quantity to retain.
- θ_2 is the nuisance variable to integrate out.
- x is the observation.

High-dimensional marginalisation is usually intractable, which is why approximate inference is needed.

Probability representations

- **Probabilistic population code (PPC).** Population firing rates encode parameters of a probability distribution. In an exponential-family PPC,

  $$p(\theta \mid x) \propto \exp(\boldsymbol{h}(\theta)^\top \boldsymbol{r}).$$

  Here θ is the latent variable, x is the observation, $\boldsymbol{h}(\theta)$ is a vector of basis functions or sufficient statistics, and $\boldsymbol{r}$ is the population response. Posterior uncertainty is commonly expressed through population gain.

- **Sampling-based code.** Instantaneous activity is treated as a draw from a distribution,

  $$\boldsymbol{r}(t) \sim p(\boldsymbol{r} \mid x).$$

  Here $\boldsymbol{r}(t)$ is population activity at time t, x is the observation, and $p(\boldsymbol{r} \mid x)$ is the target activity distribution. Uncertainty is expressed by variability across time or trials. A decoder estimates expectations from multiple samples rather than reading distribution parameters from a single activity vector.

- **Variational inference (VI).** A tractable distribution $q_{\varphi(\theta)}$ is fitted to the posterior by optimizing parameters φ. VI turns inference into optimization and is usually fast, but its approximation is limited by the chosen family for q_φ.

Exponential families and approximate inference
An exponential-family distribution has the form

$$p(x \mid \eta) = h(x) \exp(\eta^\top T(x) - A(\eta)).$$

- x is the random variable.
- η is the vector of **natural parameters**.
- $T(x)$ is the vector of **sufficient statistics**.
- $A(\eta)$ is the log-partition function.
- $h(x)$ is the base measure.

A prior and likelihood are **conjugate** when the posterior belongs to the same parametric family as the prior. Conjugacy makes some Bayesian updates analytic. PPCs use exponential-family structure because multiplying compatible distributions can become addition of their natural parameters.

Three common approximation families are:

- **Laplace approximation.** Fit a Gaussian around the posterior mode using local curvature. It is cheap but poor for strongly non-Gaussian or multimodal posteriors.
- **Variational inference.** Optimize a tractable $q_{\varphi(\theta)}$ to approximate the posterior.
- **Markov chain Monte Carlo (MCMC).** Construct a Markov chain whose stationary distribution is the posterior. It can be asymptotically exact, but finite runs can be biased by burn-in and autocorrelation.

The **Kullback–Leibler (KL) divergence** compares distributions:

$$\mathrm{KL}(q \parallel p) = \int q(\theta) \log \frac{q(\theta)}{p(\theta)} \, \mathrm{d}\theta \geq 0.$$

- $p(\theta)$ is the target distribution.
- $q(\theta)$ is the approximation.
- θ is the variable being integrated over.

The divergence is asymmetric. Minimizing $\mathrm{KL}(q \parallel p)$ is usually mode-seeking, while minimizing $\mathrm{KL}(p \parallel q)$ is usually mass-covering.

Population coding and information

- **Tuning curve.** The mean response $f_{i(\theta)}$ of neuron i as a function of stimulus θ.
- **Fisher information.** A measure of how precisely responses identify θ. Under regularity conditions, an unbiased estimator obeys the Cramér–Rao bound, $\mathrm{Var}\left(\hat{\theta}\right) \geq \frac{1}{I_{F(\theta)}}$, where $\hat{\theta}$ is the estimator and $I_{F(\theta)}$ is Fisher information.
- **Linear Fisher information.** The information accessible to a linear decoder, which is a useful proxy for what a downstream neuron can extract through weighted synaptic input.
- **Noise correlations.** Trial-to-trial covariance between neurons at fixed stimulus. Correlations aligned with the tuning gradient can cap information even as the population grows.
- **Poisson variability.** A count model in which spike-count variance equals its mean. It is a common baseline rather than a universal biological law.

Sampling dynamics and E/I circuits

- **Stationary distribution.** The distribution approached by a Markov process at long times. A neural-sampling claim requires the circuit's stationary distribution to match the intended posterior, not merely to display irregular activity.
- **Langevin dynamics.** Noisy gradient dynamics designed to approach a target stationary distribution. They provide the simplest bridge from a log-posterior landscape to recurrent stochastic dynamics.
- **Hamiltonian Monte Carlo (HMC).** A sampler that augments the represented variable with momentum, allowing longer, less diffusive moves through the target distribution. The Hamiltonian-brain account associates excitatory and inhibitory populations with complementary state variables.
- **Inhibition-stabilized network (ISN).** A recurrent E/I network whose excitatory subnetwork is unstable in isolation and stabilized by feedback inhibition.
- **Mixing time.** The time required for a chain to lose dependence on its initial state and explore the target distribution.
- **Autocorrelation time.** The time over which successive samples remain correlated. Shorter autocorrelation generally means more effective samples per unit time.
- **Transient amplification.** Temporary growth of selected activity patterns in an asymptotically stable recurrent network. It can accelerate movement through state space without requiring sustained instability.
- **Δt-stability.** In this project, robustness of the simulated or trained dynamics to the numerical integration step Δt. A purported computational regime that disappears under a smaller step may be a discretization artefact rather than a property of the continuous-time circuit.

Posterior readouts and metabolic cost

- **Maximum a posteriori (MAP) estimate.** The posterior mode, $\theta_{\mathrm{MAP}} = \arg\max_\theta p(\theta \mid x)$.
- **Posterior mean.** $\mathbb{E}[\theta \mid x]$, the Bayes-optimal point estimate under squared-error loss.
- **Posterior variance or credible interval.** The width of the posterior, representing uncertainty rather than only a best estimate.
- **Bits per spike.** Mutual information between stimulus and response divided by spike count, used as a measure of coding efficiency.
- **Coding precision.** The inverse width of the represented distribution. A system can reduce precision to save energy while retaining a similar posterior mean.
- **Metabolic cost of a spike.** The energetic cost, largely paid by ion pumps, of restoring ionic gradients after an action potential.

References

1. Pouget, Dayan & Zemel (2003), *Inference and computation with population codes* doi:10.1146/annurev.neuro.26.041002.131112
2. Ma, Beck, Latham & Pouget (2006), *Bayesian inference with probabilistic population codes* doi:10.1038/nn1790
3. Fiser, Berkes, Orbán & Lengyel (2010), *Statistically optimal perception and learning: from behavior to neural representations* doi:10.1016/j.tics.2010.01.003

4. Aitchison & Lengyel (2016), *The Hamiltonian brain: efficient probabilistic inference with excitation-inhibition networks* doi:10.1371/journal.pcbi.1005186
5. Orbán, Berkes, Fiser & Lengyel (2016), *Neural variability and sampling-based probabilistic representations in the visual cortex* doi:10.1016/j.neuron.2016.09.038
6. Echeveste, Aitchison, Hennequin & Lengyel (2020), *Cortical-like dynamics in recurrent circuits optimized for sampling-based probabilistic inference* doi:10.1038/s41593-020-0671-1
7. Padamsey, Katsanevaki, Dupuy & Rochefort (2022), *Neocortex saves energy by reducing coding precision during food scarcity* doi:10.1016/j.neuron.2021.10.024